Exact Linear Sign Representations from Canonical SQ Output Catalogs

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1 Theoretical Setup

1.1 Problem Setting

Let $\mathcal{X}$ be an arbitrary domain, possibly empty or infinite, and let $\mathcal{H} \subseteq \{+1,-1\}^{\mathcal{X}}$. For a distribution $\mathcal{D}$ on $\mathcal{X}$, a target $h \in \mathcal{H}$, and a binary predictor $g : \mathcal{X} \to \{+1,-1\}$, define

$$\mathcal{L}_{\mathcal{D},h}(g) := \Pr_{x\sim\mathcal{D}}\big[g(x)h(x) < 0\big]. \tag{1.1}$$

All query expectations and risks below are understood in the usual statistical-query sense when they arise on an actual execution. No measurability or output behavior is imposed on a prescribed response sequence that is not an actual tolerance-valid execution.

The deterministic exact dimension complexity $\mathrm{dc}(\mathcal{H})$ is the least $d \in \mathbb{N}_0$ for which there is one map $\psi : \mathcal{X} \to \mathbb{R}^d$ satisfying

$$\forall h \in \mathcal{H}\ \exists u_h \in \mathbb{R}^d\ \forall x \in \mathcal{X}, \qquad h(x)\langle u_h, \psi(x)\rangle > 0. \tag{1.2}$$

Its value is $+\infty$ when no finite d works. Thus the required representation is pointwise and exact, without a distributional exceptional set.

An unrestricted SQ query is a bounded function $q : \mathcal{X} \times \{+1,-1\} \to [-1,1]$. For fixed $(\mathcal{D}, h)$, an adaptive tolerance-τ oracle policy may choose each real reply from the reached history, but a reply a to the current query q must satisfy

$$|a - \mathbb{E}_{x\sim\mathcal{D}} q(x, h(x))| \le \tau. \tag{1.3}$$

Validity and the learner guarantee are required separately for every adversarial policy; there is no averaging over policies.

Let $(\Omega, \mathcal{F}, \mu)$ be the random-tape probability space of a learner A, and write $R \sim \mu$. The space may be arbitrary, including nonatomic. Once a complete tape $r \in \Omega$ is fixed, the learner's queries, stopping rule, and terminal binary predictor are deterministic, although a query may depend on the whole tape and all previous real replies. For a tolerance-valid policy $\mathcal{O}$, write

$$A_r^{\mathcal{O}}(\mathcal{D}, h) : \mathcal{X} \to \{+1,-1\}$$

for the terminal predictor on that actual execution.

For the tolerance τ, put

$$K := \left\lceil \frac{1}{\tau} \right\rceil, \qquad G := \left\{-1 + \frac{2j}{K} : 0 \le j \le K\right\}, \tag{1.4}$$

and fix before any instance a deterministic nearest-grid map $\rho : [-1,1] \to G$, with a fixed rule for midpoint ties. Then

$$|\rho(v) - v| \le \frac{1}{K} \le \tau \qquad (v \in [-1,1]). \tag{1.5}$$

For each $(\mathcal{D}, h)$, define the proof-only exact-center policy

$$\mathcal{O}^{\rho}_{\mathcal{D},h}(q) := \rho(\mathbb{E}_{x\sim\mathcal{D}} q(x, h(x)))\,. \tag{1.6}$$

The learner receives only the rounded reply, not the exact expectation used by the oracle.

For an ordered finite catalog $\mathcal{C} = (g_1, \dots, g_L)$, define

$$\phi_{\mathcal{C}}(x) := (g_1(x), \dots, g_L(x)) \in \{+1, -1\}^L, \qquad \Delta_L := \left\{ w \in [0,1]^L : \sum_{i=1}^{L} w_i = 1 \right\}. \tag{1.7}$$

1.2 Assumptions

Assumption 1 (SQ and catalog-family parameter regime)**.** *The parameters satisfy*

$$m \in \mathbb{N}_0, \qquad \tau \in (0,\infty), \qquad \varepsilon \in [0, 1/4), \qquad B \in [1,\infty), \qquad k \in \mathbb{N}, \quad k \ge 1.$$

The constants B and k are fixed parts of the protocol-family certificate. They are independent of $\mathcal{X}, \mathcal{H}$, the learner parameters (m, τ, ε), the instance $(\mathcal{D}, h)$, the oracle policy and replies, and the learner tape. The horizon is the fixed finite upper bound m, including $m = 0$.

Assumption 2 (Universal randomized adversarial-SQ learner)**.** *One learner A, fixed independently of $(\mathcal{D}, h)$ and of the oracle policy, makes at most m adaptive unrestricted queries of tolerance τ and returns a binary predictor. For every distribution $\mathcal{D}$ on $\mathcal{X}$, every $h \in \mathcal{H}$, and every adaptive tolerance-valid adversarial policy $\mathcal{O}$, the actual-run loss is a μ-measurable random variable and*

$$\mathbb{E}_{R\sim\mu} \mathcal{L}_{\mathcal{D},h}\big(A^{\mathcal{O}}_R(\mathcal{D}, h)\big) \le \varepsilon. \tag{1.8}$$

The expectation is only over the learner tape. The premise grants neither favorable-policy selection nor exact-expectation access to the learner.

Assumption 3 (Pre-instance canonical-policy output catalog)**.** *As part of the certified protocol, there is a specified finite ordered catalog*

$$\mathcal{C}^{\rho}_A = (g_1, \dots, g_L), \qquad 1 \le L \le B\left(1 + \frac{m}{\tau^2}\right)^k, \tag{1.9}$$

fixed after the protocol and ρ are fixed but before the current distribution, target, oracle interaction, or learner tape, such that

$$\forall \mathcal{D}\ \forall h \in \mathcal{H}\ \forall r \in \Omega, \qquad A_r^{\mathcal{O}^{\rho}_{\mathcal{D},h}}(\mathcal{D}, h) \in \mathcal{C}^{\rho}_A \tag{1.10}$$

as equality of terminal functions on $\mathcal{X}$, whenever the displayed canonical interaction is an actual tolerance-valid execution. This condition concerns only actual executions against the canonical policy. It imposes no catalog, finiteness, rank, or measurability condition on outputs under other valid policies or under prescribed tolerance-invalid response strings; the output range under other valid policies may be infinite.

1.3 Formalized Goal

Under Assumptions 1, 2, and 3, the objective is to prove that the deterministic common map

$$\phi := \phi_{\mathcal{C}_A^\rho} : \mathcal{X} \to \mathbb{R}^L, \qquad \phi(x) = (g_1(x), \ldots, g_L(x)),$$

is fixed independently of the current distribution, target, valid oracle policy, and learner tape, and satisfies

$$\forall h \in \mathcal{H}\ \exists w_h \in \Delta_L\ \forall x \in \mathcal{X}, \qquad h(x)\langle w_h, \phi(x)\rangle \geq 1 - 2\varepsilon > \frac{1}{2} > 0. \tag{1.11}$$

Each w_h may depend on h, but not on $\mathcal{D}$, the oracle policy, or the learner tape. Consequently, the desired exact deterministic bound is

$$\mathrm{dc}(\mathcal{H}) \leq L \leq B\left(1 + \frac{m}{\tau^2}\right)^k. \tag{1.12}$$

This is a conditional result under Assumption 3. It does not derive that catalog from (m, τ) alone and does not resolve the unconditional open question of obtaining a linear $O(m/\tau^2)$ dimension bound from the unrestricted SQ guarantee alone.

2 Preliminaries

For the ordered catalog in Assumption 3, write

$$\rho_\varepsilon := 1 - 2\varepsilon, \qquad \phi(x) := (g_1(x), \ldots, g_L(x)), \qquad \Delta_L := \left\{ w \in [0,1]^L : \sum_{i=1}^{L} w_i = 1 \right\}. \tag{2.1}$$

The scalar ρ_ε names the exact margin appearing in the main theorem. The map ϕ is the catalog coordinate map from (1.7), and Δ_L is the set of catalog mixtures used for the target-dependent weights. These objects are derived directly from the formalized setting; no admissibility event, selector, stability condition, or additional assumption is encoded in this notation.

3 Main Theorem

Theorem 3.1 (Conditional canonical-catalog representation)**.** *Under Assumptions 1, 2, and 3, the one deterministic map $\phi : \mathcal{X} \to \mathbb{R}^L$ in (2.1) is fixed independently of the current distribution, target, valid oracle policy, and learner tape, and*

$$\forall h \in \mathcal{H}\ \exists w_h \in \Delta_L\ \forall x \in \mathcal{X}, \qquad h(x)\langle w_h, \phi(x)\rangle \geq 1 - 2\varepsilon > \frac{1}{2} > 0. \tag{3.1}$$

Only h may affect w_h. Consequently,

$$\mathrm{dc}(\mathcal{H}) \leq L \leq B\left(1 + \frac{m}{\tau^2}\right)^k. \tag{3.2}$$

If $\mathcal{H} = \varnothing$ or $\mathcal{X} = \varnothing$, then in fact $\mathrm{dc}(\mathcal{H}) = 0$.

The conclusion is deterministic and pointwise, with no exceptional set. It uses the fixed finite horizon m, including $m = 0$, and has no hidden constant: all dependence on $m, \tau, \varepsilon, L, B, k$ is displayed, while B, k are the fixed protocol-family constants described in Assumption 1. The result is conditional on the canonical-policy-only catalog in Assumption 3; it asserts no catalog property for other valid policies.

4 Proof Sketch

First, Lemma A.1 verifies the additive SQ condition separately at every reached query of the canonical rounded policy. Proposition A.1 may therefore invoke the catalog assumption on an actual execution, and only on that execution.

For fixed $(\mathcal{D}, h)$, Lemma A.2 considers only the measurable scalar actual-run loss. Its finite range has an attained minimum no larger than its expectation, giving an actually occurring catalog function of risk at most ε, without a measurable catalog selector. Proposition A.2 converts this risk exactly to correlation at least ρ_ε.

For a nonempty finite $F \subseteq \mathcal{X}$, instantiate that correlation statement at every finitely supported law on F. The resulting finite matrix game has value at least ρ_ε. Sion's minimax theorem (Sion, 1958, Theorem 3.4), applied only after all of its finite-dimensional hypotheses are checked, reverses the two optimization orders. Proposition A.3 then gives one catalog mixture meeting every margin constraint on F; the empty set is handled separately by Lemma A.6.

For fixed h, each pointwise margin constraint is a closed subset of the single compact simplex Δ_L. Finite feasibility gives the finite-intersection property, and the open-cover form of compactness yields the global weight in Proposition A.4. The coordinate identity in Lemma A.11 then turns that mixture into the exact Euclidean score. Propositions A.5 and A.6 finish the strict-sign and dimension conclusions without changing the margin or introducing a hidden factor.

References

SION, M. (1958). On general minimax theorems. *Pacific Journal of Mathematics* **8** 171–176. 4, 11

A Proof Details

A.1 Legality of the Canonical Rounded Execution

Lemma A.1 (Pathwise legality of the canonical rounded policy)**.** *Under Assumptions 1 and 2, fix a distribution $\mathcal{D}$, a target $h \in \mathcal{H}$, and a tape $r \in \Omega$. Generate the interaction recursively by replying to each actually reached bounded query q_t with*

$$v_t := \mathbb{E}_{x\sim\mathcal{D}} q_t(x, h(x)), \qquad a_t := \rho(v_t).$$

Then every reached reply satisfies

$$|a_t - v_t| \le \frac{1}{K} \le \tau, \qquad K = \left\lceil \frac{1}{\tau} \right\rceil.$$

Consequently, the recursion is an actual tolerance-valid execution of some length $T \le m$. This includes early stopping and the empty execution when $m = 0$, for every finite $\tau > 0$, every fixed midpoint rule, and every tape r, without a condition on the tape law.

Proof. Because $\tau > 0$, the integer $K = \lceil 1/\tau \rceil$ is finite and positive. The grid in (1.4) has adjacent spacing $2/K$ and covers $[-1, 1]$. A point between two consecutive grid points is within half that spacing of at least one endpoint. Thus, for every $v \in [-1, 1]$,

$$|\rho(v) - v| \le \frac{1}{K}. \tag{A.1}$$

At a midpoint both adjacent points are at distance $1/K$, so the fixed tie rule preserves the bound. Moreover,

$$K = \left\lceil \frac{1}{\tau} \right\rceil \ge \frac{1}{\tau} > 0 \quad \Longrightarrow \quad \frac{1}{K} \le \tau. \tag{A.2}$$

This also covers $\tau \ge 1$: then $K = 1$, the grid is $\{-1, 1\}$, and its covering radius is $1 \le \tau$.

Fix the chosen $(\mathcal{D}, h, r)$. Once the tape is fixed, the learner's next action is deterministic given the replies actually received. Begin with the empty reached history. If the learner stops, the interaction is complete. Otherwise, it issues a first query $q_1 : \mathcal{X} \times \{+1, -1\} \to [-1, 1]$. Pointwise boundedness gives

$$-1 \le q_1(x, h(x)) \le 1,$$

and hence $v_1 \in [-1, 1]$. Equations (A.1)–(A.2) give $|a_1 - v_1| \le \tau$, exactly the additive condition (1.3).

For the induction step, suppose the first $t - 1$ canonical replies form the reached valid prefix. That prefix and the fixed tape determine whether the learner stops or issues a next bounded query q_t. In the latter case, the same pointwise bound gives $v_t \in [-1, 1]$, regardless of how q_t depends on the tape and earlier replies. Therefore

$$|a_t - v_t| = |\rho(v_t) - v_t| \le \frac{1}{K} \le \tau, \tag{A.3}$$

and appending a_t yields the next valid reached prefix. The induction continues only until the learner stops, and Assumption 2 bounds the reached length by $T \le m$. When $T \ge 1$, the pathwise conclusion may equivalently be written

$$\max_{1 \le t \le T} |a_t - v_t| \le \tau.$$

No bound on $\sum_t |a_t - v_t|$ is required or asserted. Adaptive changes of later queries therefore create no accumulated response defect.

If the learner stops after a valid prefix, no unreached query requires a reply. If $m = 0$, no query can be reached, so the empty interaction is the actual tolerance-valid execution by vacuity. The argument is pointwise in r and uses neither μ, atoms, nor finite support of the tape space. The oracle uses v_t only to compute a_t; the learner receives a_t, not v_t. Finally, the recursion follows only the history generated by these canonical replies. It does not replay a prescribed tolerance-invalid string or assign such a string a terminal output. ☐

Proposition A.1 (Exact membership on the actual canonical execution)**.** *Under Assumptions 1, 2, and 3, together with Lemma A.1, for every distribution $\mathcal{D}$, target $h \in \mathcal{H}$, and tape $r \in \Omega$,*

$$A_r^{\mathcal{O}^{\rho}_{\mathcal{D},h}}(\mathcal{D}, h) \in \mathcal{C}^{\rho}_A = (g_1, \ldots, g_L)$$

as equality of functions on $\mathcal{X}$. The catalog is the same ordered catalog fixed before the instance. The statement includes the no-query execution and makes no claim about another valid policy or a prescribed tolerance-invalid response string.

Proof. Fix $(\mathcal{D}, h, r)$. By Lemma A.1, the recursively generated interaction against $\mathcal{O}^{\rho}_{\mathcal{D},h}$ is an actual tolerance-valid canonical execution, including early stopping and the empty execution. The antecedent of Assumption 3 is therefore met. That assumption supplies an index $i \in \{1, \ldots, L\}$, not necessarily unique when catalog entries repeat, such that

$$A_r^{\mathcal{O}^{\rho}_{\mathcal{D},h}}(\mathcal{D}, h)(x) = g_i(x) \qquad \text{for every } x \in \mathcal{X}. \tag{A.4}$$

This is zero-residual equality of terminal functions, rather than approximate membership. Since the fixed triple was arbitrary, the conclusion holds for all instances and tapes. No measurable selection of i as a function of r is made, so the argument is unchanged for an arbitrary or nonatomic tape space.

Together with Lemma A.1, this proves that every canonical interaction used below is both an actual valid execution and catalog-valued. The scope remains canonical-policy-only: no conclusion is drawn for another valid policy, a synthetic invalid transcript, or an all-policy output range. ☐

Conclusion of the canonical-execution argument. Let $(\mathcal{D}, h)$ be arbitrary. Lemma A.1 applies pathwise to every tape: each reached response satisfies $|a_t - v_t| \le 1/K \le \tau$, so the adaptive recursion is an actual valid execution of length at most m. This includes early stopping, the actual no-query execution at $m = 0$, fixed midpoint ties, every $\tau > 0$, and an arbitrary tape space. Proposition A.1 then gives

$$\forall r \in \Omega, \qquad A_r^{\mathcal{O}^{\rho}_{\mathcal{D},h}}(\mathcal{D}, h) \in \mathcal{C}^{\rho}_A$$

as equality of functions on $\mathcal{X}$. Since $(\mathcal{D}, h)$ was arbitrary, these conclusions provide exactly the actual canonical executions and fixed-catalog membership used later, without defining an output on an invalid string or extending the catalog to another policy. ☐

A.2 Selector-Free Extraction of a Correlated Catalog Function

Lemma A.2 (Attainment of a low-risk actual catalog output)**.** *Under Assumptions 1, 2, and 3, together with Lemma A.1 and Proposition A.1, fix a distribution $\mathcal{D}$ and target $h \in \mathcal{H}$. For each $r \in \Omega$, set*

$$G_r := A_r^{\mathcal{O}^{\rho}_{\mathcal{D},h}}(\mathcal{D}, h), \qquad Z(r) := \mathcal{L}_{\mathcal{D},h}(G_r).$$

Then $Z : \Omega \to [0,1]$ is μ-measurable, its range is nonempty and finite, and there are $r_* \in \Omega$ and $i_* \in \{1, \dots, L\}$ such that

$$G_{r_*} = g_{i_*} \quad \text{on } \mathcal{X}, \qquad \mathcal{L}_{\mathcal{D},h}(g_{i_*}) = Z(r_*) \le \varepsilon.$$

Occurrence means existence of the tape r_* on the actual canonical execution; no atom or positive-mass output fiber is required.

Proof. Lemma A.1 proves that the canonical policy is tolerance-valid along the actual execution for every tape. Instantiating Assumption 2 at this one proved-valid policy and the fixed $(\mathcal{D}, h)$ gives

$$Z : \Omega \to [0,1] \text{ measurable}, \qquad \mathbb{E}_{R\sim\mu} Z(R) \le \varepsilon. \tag{A.5}$$

No random function, catalog coordinate, or output index is declared measurable in obtaining (A.5).

Proposition A.1 gives, separately for each tape,

$$G_r \in \{g_1, \dots, g_L\}$$

as exact equality of functions. Consequently, the actual terminal function has at most L distinct values, and the scalar range

$$S := Z(\Omega)$$

has at most L distinct real values. This is a set-theoretic statement, not a pushforward probability law or measurability claim for an index map. Because $\mu(\Omega) = 1$, the set Ω, and hence S, is nonempty. Thus $S \subseteq [0,1]$ is nonempty and finite.

Let $s_* := \min S$. By the definition of the range, some $r_* \in \Omega$ satisfies $Z(r_*) = s_*$. Pointwise, $Z \ge s_*$, so positivity of the integral applied to $Z - s_*$ gives

$$s_* = \int_\Omega s_* \, d\mu \le \int_\Omega Z \, d\mu = \mathbb{E}_{R\sim\mu} Z(R) \le \varepsilon. \tag{A.6}$$

Only the bounded measurable scalar Z is integrated; no output function, index, fiber indicator, or catalog-coordinate loss is integrated.

Apply Proposition A.1 once at r_*. It gives an $i_* \in \{1, \dots, L\}$ such that $G_{r_*} = g_{i_*}$ pointwise. Hence

$$\mathcal{L}_{\mathcal{D},h}(g_{i_*}) = \mathcal{L}_{\mathcal{D},h}(G_{r_*}) = Z(r_*) = s_* \le \varepsilon. \tag{A.7}$$

The choice is a single existential choice after $(\mathcal{D}, h, r_*)$ is fixed. Duplicate entries need not have a unique index, and no selector over tapes, distributions, or targets is constructed.

The proof neither uses an atom of μ, finite tape support, nor positive mass or measurability of an output fiber. It does not require the singleton $\{r_*\}$ to be measurable. For $L = 1$, the scalar range is a singleton; duplicates only reduce its cardinality. For $m = 0$, the actual empty execution supplied by Lemma A.1 precedes (A.5). Every $\tau > 0$ remains covered. If $\varepsilon = 0$, then $0 \le s_* \le 0$, so the occurring catalog function has risk zero. ☐

Proposition A.2 (Exact same-function conversion from risk to correlation)**.** *Under Assumptions 1, 2, and 3, together with Lemma A.2, for every distribution $\mathcal{D}$ and target $h \in \mathcal{H}$, some actually occurring catalog function g_i satisfies*

$$\mathbb{E}_{x\sim\mathcal{D}}[h(x)g_i(x)] = 1 - 2\mathcal{L}_{\mathcal{D},h}(g_i) \ge 1 - 2\varepsilon = \rho_\varepsilon. \tag{A.8}$$

The risk and correlation concern the same function, target, and distribution.

Proof. Fix $(\mathcal{D}, h)$, and let g_{i_*} be the function supplied by Lemma A.2. Since $h(x), g_{i_*}(x) \in \{+1, -1\}$, their product is $+1$ when they agree and -1 when they disagree. Pointwise,

$$h(x)g_{i_*}(x) = 1 - 2\mathbf{1}\{g_{i_*}(x)h(x) < 0\}. \tag{A.9}$$

The indicator is exactly the event in (1.1). Taking the $\mathcal{D}$-expectation in (A.9) gives

$$\mathbb{E}_{x\sim\mathcal{D}}[h(x)g_{i_*}(x)] = 1 - 2 \Pr_{x\sim\mathcal{D}}[g_{i_*}(x)h(x) < 0] = 1 - 2\mathcal{L}_{\mathcal{D},h}(g_{i_*}). \tag{A.10}$$

By (A.7), the risk is at most ε. Multiplication by -2 reverses the inequality, so

$$1 - 2\mathcal{L}_{\mathcal{D},h}(g_{i_*}) \geq 1 - 2\varepsilon = \rho_\varepsilon.$$

There is no approximation, rounding term, change of function, or distributional transfer. At $\varepsilon = 0$, the same argument gives correlation exactly one. Since the fixed pair was arbitrary, this proves the stated per-distribution catalog response without a measurable selector, output law, positive-mass fiber, or output from another policy. □

Conclusion of the scalar-loss extraction argument. Fix an arbitrary $(\mathcal{D}, h)$. Lemma A.1 makes the canonical policy valid, and Proposition A.1 makes its terminal range finite. Lemma A.2 then uses only the measurable scalar actual-run loss to produce an actually occurring g_{i_*} with $\mathcal{L}_{\mathcal{D},h}(g_{i_*}) \leq \varepsilon$. Proposition A.2 applies the pointwise binary identity to that same function and gives

$$\mathbb{E}_{x\sim\mathcal{D}}[h(x)g_{i_*}(x)] \geq 1 - 2\varepsilon = \rho_\varepsilon.$$

Arbitrariness of the pair yields the required $\forall(\mathcal{D}, h)\, \exists i$ interface. No selector, fiber probability, finite-support tape reduction, irrelevant catalog coordinate, other policy, or invalid transcript enters the conclusion. □

A.3 Finite-Distribution Minimax and Simultaneous Separation

For a nonempty finite set F, write

$$\Delta_F := \left\{ p \in [0,1]^F : \sum_{x\in F} p_x = 1 \right\}, \qquad [L] := \{1, \ldots, L\}.$$

Let e_i denote the i-th vertex of Δ_L, and let d_x denote the point-mass vertex of Δ_F.

Lemma A.3 (Finite-support instances lower-bound the distribution game)**.** *Under Assumptions 1, 2, and 3, together with Proposition A.2, fix $h \in \mathcal{H}$ and a nonempty finite $F \subseteq \mathcal{X}$. Define*

$$A \in \mathbb{R}^{F\times L}, \qquad A_{xi} := h(x)g_i(x). \tag{A.11}$$

Then every $p \in \Delta_F$ satisfies

$$\max_{i\in[L]} p^T A e_i \geq \rho_\varepsilon,$$

and the minimum exists and obeys

$$\min_{p\in\Delta_F} \max_{i\in[L]} p^T A e_i \geq \rho_\varepsilon. \tag{A.12}$$

Proof. Fix $p \in \Delta_F$. Define the finitely supported probability law

$$\mathcal{D}_p := \sum_{x \in F} p_x \delta_x, \qquad \mathbb{E}_{z \sim \mathcal{D}_p}[u(z)] := \sum_{x \in F} p_x u(x) \tag{A.13}$$

for every pointwise function used below. Coordinates with $p_x = 0$ cause no difficulty; the support is merely contained in F. This finite-sum definition introduces no singleton-measurability condition.

Proposition A.2 applies because it ranges over every distribution, not only full-support or nonatomic laws. It gives an index $i(p) \in [L]$ such that

$$\begin{aligned} p^T A e_{i(p)} &= \sum_{x \in F} p_x A_{x,i(p)} \\ &= \sum_{x \in F} p_x h(x) g_{i(p)}(x) \\ &= \mathbb{E}_{x \sim \mathcal{D}_p}[h(x) g_{i(p)}(x)] \\ &\geq \rho_\varepsilon. \end{aligned} \tag{A.14}$$

Every expectation here is the displayed finite sum, so this proof instance adds no topology or measurability condition on F, p, or the catalog coordinates. It also does not narrow the universal premise to finite-support distributions.

Since $i(p)$ is among the finitely many maximized indices,

$$\max_{i \in [L]} p^T A e_i \geq p^T A e_{i(p)} \geq \rho_\varepsilon. \tag{A.15}$$

No common index for different p's is asserted. The map $p \mapsto \max_i p^T A e_i$ is a finite maximum of continuous affine functions. The nonempty simplex Δ_F is closed and bounded in a finite-dimensional Euclidean space, hence compact, so this continuous map has a minimizer p_*. Evaluating (A.15) at p_* gives

$$\min_{p \in \Delta_F} \max_{i \in [L]} p^T A e_i = \max_{i \in [L]} p_*^T A e_i \geq \rho_\varepsilon.$$

□

Lemma A.4 (Catalog-simplex maximization occurs at a column vertex)**.** *Under Assumption 3 and the matrix construction (A.11), if $F \neq \varnothing$ is finite and $p \in \Delta_F$, then*

$$\max_{w \in \Delta_L} p^T A w = \max_{i \in [L]} p^T A e_i, \tag{A.16}$$

and both sides are attained.

Proof. Fix p and put $c_i := p^T A e_i$. For every $w \in \Delta_L$,

$$\begin{aligned} p^T A w = \sum_{i=1}^{L} w_i p^T A e_i &= \sum_{i=1}^{L} w_i c_i \\ &\leq \sum_{i=1}^{L} w_i \max_{j \in [L]} c_j = \max_{j \in [L]} c_j. \end{aligned} \tag{A.17}$$

Because $[L]$ is finite and nonempty, choose $i_* \in \arg\max_i c_i$. The vertex $e_{i_*} \in \Delta_L$ gives

$$p^T A e_{i_*} = c_{i_*} = \max_{i \in [L]} c_i. \tag{A.18}$$

Equations (A.17) and (A.18) prove the identity and attainment. Zero coordinates of w and duplicate columns are allowed. □

Lemma A.5 (Distribution-simplex minimization occurs at a point mass)**.** *Under the matrix construction (A.11), if $F \neq \varnothing$ is finite and $w \in \Delta_L$, then*

$$\min_{p \in \Delta_F} p^T A w = \min_{x \in F} (Aw)_x, \tag{A.19}$$

and both sides are attained.

Proof. For every $p \in \Delta_F$,

$$\begin{aligned} p^T A w &= \sum_{x \in F} p_x (Aw)_x \\ &\geq \sum_{x \in F} p_x \min_{y \in F} (Aw)_y = \min_{y \in F} (Aw)_y. \end{aligned} \tag{A.20}$$

Since F is finite and nonempty, choose $x_* \in \arg\min_{x \in F} (Aw)_x$. Its point mass $d_{x_*} \in \Delta_F$ satisfies

$$d_{x_*}^T A w = (Aw)_{x_*} = \min_{x \in F} (Aw)_x. \tag{A.21}$$

This proves the identity and attainment. In particular, boundary points of Δ_F, whose other coordinates vanish, are included. □

Proposition A.3 (Exact finite minimax separator)**.** *Under Assumptions 1, 2, and 3, together with Proposition A.2 and Lemmas A.3, A.4, and A.5, fix $h \in \mathcal{H}$ and a nonempty finite $F \subseteq \mathcal{X}$, and let $A_{xi} = h(x) g_i(x)$. Then*

$$\min_{p \in \Delta_F} \max_{i \in [L]} p^T A e_i = \max_{w \in \Delta_L} \min_{x \in F} (Aw)_x \geq \rho_\varepsilon. \tag{A.22}$$

The maximum on the right is attained by some $w_{h,F} \in \Delta_L$, and

$$\forall x \in F, \qquad h(x) \sum_{i=1}^{L} w_{h,F,i} g_i(x) \geq \rho_\varepsilon. \tag{A.23}$$

Proof. We first record the cited minimax statement in the orientation used here. Sion's Theorem 3.4 (Sion, 1958) says that if M is the maximizing domain, N is the minimizing domain, both are nonempty compact convex sets, and a real payoff $f : M \times N \to \mathbb{R}$ is upper semicontinuous and quasi-concave in its first variable and lower semicontinuous and quasi-convex in its second, then

$$\sup_{w \in M} \inf_{p \in N} f(w, p) = \inf_{p \in N} \sup_{w \in M} f(w, p). \tag{A.24}$$

Take $M = \Delta_L$, $N = \Delta_F$, and $f(w, p) := p^T A w$. The conditions $L \geq 1$ and $F \neq \varnothing$ make the simplices nonempty. Each simplex is convex, closed, and bounded in a finite-dimensional Euclidean space, hence compact. For fixed p, the function $w \mapsto p^T A w$ is real-valued, continuous, and affine, hence quasi-concave. For fixed w, the function $p \mapsto p^T A w$ is real-valued, continuous, and affine, hence quasi-convex. Thus the variables, topology, convexity, semicontinuity, and quasi-concavity/quasi-convexity conventions all match (A.24). The theorem gives only

$$\sup_{w \in \Delta_L} \inf_{p \in \Delta_F} p^T A w = \inf_{p \in \Delta_F} \sup_{w \in \Delta_L} p^T A w. \tag{A.25}$$

Attainment is checked separately. By Lemma A.5, the left inner infimum is the function $w \mapsto \min_{x \in F} (Aw)_x$, a finite minimum of continuous affine functions and therefore continuous. Compactness of the nonempty Δ_L gives a maximizer $w_{h,F}$. By Lemma A.4, the right inner supremum is $p \mapsto \max_{i \in [L]} p^T A e_i$, a finite maximum of continuous affine functions. Its minimum on compact Δ_F is attained, as also shown in Lemma A.3. Hence the suprema and infima in (A.25) may be written as maxima and minima.

Substituting the two independently proved vertex identities and then using Lemma A.3 gives

$$\begin{aligned}\max_{w\in\Delta_L}\min_{x\in F}(Aw)_x &= \max_{w\in\Delta_L}\min_{p\in\Delta_F} p^T Aw \\ &= \min_{p\in\Delta_F}\max_{w\in\Delta_L} p^T Aw \\ &= \min_{p\in\Delta_F}\max_{i\in[L]} p^T Ae_i \\ &\geq \rho_\varepsilon. \end{aligned}\tag{A.26}$$

Reading the equality in the opposite direction gives exactly the orientation in (A.22): the distribution player is the outer minimizer and the catalog-mixture player is the outer maximizer. Sion's theorem supplies the middle equality only; Lemma A.3 supplies positivity, and Lemmas A.4 and A.5 supply the endpoint identities.

For the maximizer $w_{h,F}$, (A.26) gives $\min_{x\in F}(Aw_{h,F})_x \geq \rho_\varepsilon$. Therefore, for each $x \in F$,

$$(Aw_{h,F})_x = \sum_{i=1}^{L} A_{xi} w_{h,F,i} = h(x)\sum_{i=1}^{L} w_{h,F,i} g_i(x) \geq \rho_\varepsilon. \tag{A.27}$$

The matrix payoff and the exported pointwise score are identical; there is no surrogate object or margin loss.

For a singleton F, Δ_F contains only its point mass. For $L = 1$, $\Delta_L = \{e_1\}$. Duplicate columns preserve every calculation, and zero weights are permitted. If $\varepsilon = 0$, then $\rho_\varepsilon = 1$. The game value is at least one, while every entry of A belongs to $\{-1, +1\}$, so every convex-combination payoff is at most one. Hence the game value and every finite-set margin are exactly one. □

Lemma A.6 (The empty finite restriction is vacuously feasible)**.** *Under Assumptions 1 and 3, if $h \in \mathcal{H}$ and $F = \varnothing$, then $e_1 \in \Delta_L$ and*

$$\forall x \in F, \qquad h(x)\sum_{i=1}^{L}(e_1)_i g_i(x) \geq \rho_\varepsilon$$

holds vacuously.

Proof. Assumption 3 gives $L \geq 1$, so the first simplex vertex e_1 exists and belongs to Δ_L. There is no $x \in \varnothing$ at which the inequality can fail. This branch defines no probability simplex or probability distribution indexed by the empty set, takes no minimum over an empty index set, and does not invoke minimax. Together with Proposition A.3, it proves exact ρ_ε-feasibility on every finite restriction, without asserting compatible choices for different finite sets. □

Conclusion of the finite-separation argument. Fix $h \in \mathcal{H}$ and finite $F \subseteq \mathcal{X}$. If $F = \varnothing$, Lemma A.6 supplies a weight in Δ_L and the constraints are vacuous. If $F \neq \varnothing$, Lemma A.3 gives the positive distribution-game value, while Lemmas A.4 and A.5 prove the two simplex identities. Proposition A.3 combines these facts with the fully checked Sion equality and independent attainment to obtain

$$\min_{p\in\Delta_F}\max_{i\in[L]} p^T Ae_i = \max_{w\in\Delta_L}\min_{x\in F}(Aw)_x \geq \rho_\varepsilon, \qquad A_{xi} = h(x)g_i(x),$$

and a maximizer $w_{h,F}$ satisfying

$$h(x)\sum_{i=1}^{L} w_{h,F,i} g_i(x) \geq \rho_\varepsilon \qquad (x \in F).$$

Since h and F were arbitrary, both branches give exact finite feasibility. No compatibility between weights for different F's is asserted or needed. $\square$

A.4 Fixed-Simplex Globalization

Lemma A.7 (The fixed catalog simplex is nonempty and compact)**.** *Under Assumption 3, the simplex*

$$\Delta_L = \left\{ w \in [0,1]^L : \sum_{i=1}^{L} w_i = 1 \right\} \subseteq \mathbb{R}^L$$

is nonempty and compact.

Proof. Assumption 3 gives a finite integer $L \geq 1$. Hence $e_1 = (1, 0, \ldots, 0)$ belongs to Δ_L, proving nonemptiness.

For each coordinate, the condition $0 \leq w_i \leq 1$ defines a closed subset of $\mathbb{R}^L$. The map $w \mapsto \sum_{i=1}^{L} w_i$ is continuous, so the affine hyperplane on which the sum equals one is closed. Their finite intersection is Δ_L, which is therefore closed. Moreover,

$$\|w\|_2^2 = \sum_{i=1}^{L} w_i^2 \leq \sum_{i=1}^{L} w_i = 1 \qquad (w \in \Delta_L), \tag{A.28}$$

where $w_i^2 \leq w_i$ follows from $0 \leq w_i \leq 1$. Thus the simplex is bounded. Since L is finite, the Heine–Borel theorem gives compactness. For $L = 1$, this reads $\Delta_1 = \{(1)\}$. $\square$

Lemma A.8 (Each pointwise margin constraint is relatively closed)**.** *Under Assumptions 1 and 3, fix $h \in \mathcal{H}$ and $x \in \mathcal{X}$, and define*

$$\ell_{h,x}(w) := h(x) \sum_{i=1}^{L} w_i g_i(x), \qquad C_{h,x} := \{ w \in \Delta_L : \ell_{h,x}(w) \geq \rho_\varepsilon \}. \tag{A.29}$$

Then $\ell_{h,x} : \mathbb{R}^L \to \mathbb{R}$ is a continuous linear functional and $C_{h,x}$ is relatively closed in Δ_L.

Proof. The coefficients $h(x)g_i(x)$ lie in $\{-1, +1\}$. For $u, v \in \mathbb{R}^L$,

$$\begin{aligned} |\ell_{h,x}(u) - \ell_{h,x}(v)| &= \left| \sum_{i=1}^{L} h(x) g_i(x)(u_i - v_i) \right| \\ &\leq \sum_{i=1}^{L} |u_i - v_i| \leq L \|u - v\|_2. \end{aligned} \tag{A.30}$$

Thus $\ell_{h,x}$ is Lipschitz and continuous. The ray $[\rho_\varepsilon, \infty)$ is closed in $\mathbb{R}$, and

$$C_{h,x} = \Delta_L \cap \ell_{h,x}^{-1}([\rho_\varepsilon, \infty)) = \big(\ell_{h,x}|_{\Delta_L}\big)^{-1}([\rho_\varepsilon, \infty)). \tag{A.31}$$

Hence $C_{h,x}$ is closed in the subspace topology of Δ_L. The proof uses x only as an index and assumes no topology or sigma-algebra on $\mathcal{X}$. Duplicate catalog functions do not affect the calculation. $\square$

Lemma A.9 (Finite separators give the finite-intersection property)**.** *Under Assumptions 1, 2, and 3, together with Proposition A.3, Lemma A.6, and Lemma A.7, fix $h \in \mathcal{H}$ and define $C_{h,x}$ by (A.29). Then every finite $F \subseteq \mathcal{X}$, including $F = \varnothing$, satisfies*

$$\bigcap_{x \in F} C_{h,x} \neq \varnothing, \tag{A.32}$$

where the empty-subfamily intersection is Δ_L.

Proof. If $F \neq \varnothing$, Proposition A.3 gives $w_{h,F} \in \Delta_L$ such that

$$h(x) \sum_{i=1}^{L} w_{h,F,i} g_i(x) \geq \rho_\varepsilon \qquad (x \in F).$$

By (A.29), this is precisely

$$w_{h,F} \in \bigcap_{x \in F} C_{h,x}.$$

If $F = \varnothing$, the intersection of the empty subfamily of subsets of Δ_L is, by definition, Δ_L. It contains e_1 by Lemma A.6, while Lemma A.7 independently proves nonemptiness. A singleton subfamily is covered by the nonempty branch. The witness may depend arbitrarily on F; no compatible family $F \mapsto w_{h,F}$ is selected or assumed. □

Lemma A.10 (Closed-set finite-intersection principle)**.** *Let Y be a compact topological space, let J be an arbitrary index set, and let $\{D_j : j \in J\}$ be a family of subsets closed in Y. Adopt the convention $\bigcap_{j \in \varnothing} D_j = Y$. If*

$$\bigcap_{j \in E} D_j \neq \varnothing \qquad \textit{for every finite } E \subseteq J,$$

including $E = \varnothing$, then

$$\bigcap_{j \in J} D_j \neq \varnothing.$$

Proof. The hypothesis for $E = \varnothing$ says $Y \neq \varnothing$. If $J = \varnothing$, the total intersection is Y, so the conclusion is immediate. Otherwise, suppose for contradiction that

$$\bigcap_{j \in J} D_j = \varnothing. \tag{A.33}$$

For $j \in J$, let $U_j := Y \setminus D_j$. Each U_j is open in Y. Taking complements in (A.33) gives

$$Y = \bigcup_{j \in J} U_j,$$

so the U_j's form an open cover. Compactness yields a finite $E \subseteq J$ such that

$$Y = \bigcup_{j \in E} U_j.$$

Taking complements inside Y and applying De Morgan's law gives

$$\bigcap_{j \in E} D_j = Y \setminus \bigcup_{j \in E} U_j = \varnothing, \tag{A.34}$$

contradicting the assumed nonemptiness of every finite intersection. The argument applies to every cardinality of J and uses neither a sequence nor sequential compactness. □

Proposition A.4 (One fixed-simplex separator works on the arbitrary domain)**.** *Under Assumptions 1, 2, and 3, together with Proposition A.3, Lemma A.6, and Lemmas A.7, A.8, A.9, and A.10, for every $h \in \mathcal{H}$ there is a $w_h \in \Delta_L$ such that*

$$\forall x \in \mathcal{X}, \qquad h(x) \sum_{i=1}^{L} w_{h,i} g_i(x) \geq \rho_\varepsilon = 1 - 2\varepsilon. \tag{A.35}$$

Proof. Fix $h \in \mathcal{H}$. By Lemma A.7, the space $Y := \Delta_L$ is nonempty and compact. By Lemma A.8, every $D_x := C_{h,x}$, $x \in \mathcal{X}$, is closed in this same space. By Lemma A.9,

$$\bigcap_{x \in F} D_x \neq \varnothing \qquad \text{for every finite } F \subseteq \mathcal{X},$$

including the empty subfamily. Lemma A.10 therefore applies with $J = \mathcal{X}$, and gives

$$\bigcap_{x \in \mathcal{X}} C_{h,x} \neq \varnothing. \tag{A.36}$$

Choose w_h from this intersection. For each x, membership in $C_{h,x}$ unfolds exactly to (A.35). The finite and global inequalities use the identical catalog, target, simplex, point evaluation, and threshold; the transfer has zero residual.

If $\mathcal{X} = \varnothing$, the total intersection is the empty-family intersection Δ_L, and e_1 is an explicit choice; the pointwise conclusion is vacuous. If $\mathcal{X}$ is finite, countably infinite, or uncountable, the same arbitrary-index-set lemma applies without a topology, ordering, sigma-algebra, or countability condition on $\mathcal{X}$. If $L = 1$, the simplex is $\{(1)\}$, and finite feasibility puts this sole weight in every finite intersection. Duplicate catalog functions alter neither the ambient simplex, closedness, nor finite feasibility.

If $\varepsilon = 0$, every constraint has threshold one. For any $w \in \Delta_L$, the value $h(x) \sum_i w_i g_i(x)$ is a convex combination of numbers in $\{-1, +1\}$, so it is at most one. Thus every nonvacuous global margin is exactly one, not one minus a compactness residual. At no point is a compatible family of finite witnesses selected: all constraints live in the single fixed simplex, and no varying simplex, limiting simplex, subsequence, net, or topology on the index set is introduced. □

Conclusion of the fixed-simplex globalization argument. For an arbitrary h, Lemma A.7 supplies the one nonempty compact ambient simplex, Lemma A.8 supplies the exact closed pointwise constraints, and Lemma A.9 translates Propositions A.3 and Lemma A.6 into their finite-intersection property. Lemma A.10 then yields the total intersection, and Proposition A.4 gives one w_h satisfying (A.35) on the entire arbitrary domain. Since h was arbitrary, the quantifiers are $\forall h \, \exists w_h \, \forall x$, and the threshold remains ρ_ε with zero loss. □

A.5 Exact Coordinates, Signs, and Dimension Closure

Lemma A.11 (Exact pre-instance catalog coordinates)**.** *Under Assumption 3 and Proposition A.1, let $\mathcal{C}_A^\rho = (g_1, \ldots, g_L)$ be the exact ordered catalog and define*

$$\phi : \mathcal{X} \to \mathbb{R}^L, \qquad \phi(x) := (g_1(x), \ldots, g_L(x)). \tag{A.37}$$

Then ϕ is fixed before and independently of the current distribution, target, oracle policy, and learner tape. For every $w = (w_1, \ldots, w_L) \in \mathbb{R}^L$ and $x \in \mathcal{X}$,

$$\langle w, \phi(x) \rangle = \sum_{i=1}^{L} w_i g_i(x). \tag{A.38}$$

The identity is exact for $L = 1$, repeated catalog functions, and arbitrary finite or infinite $\mathcal{X}$.

Proof. Assumption 3 fixes the ordered list $(g_1, \ldots, g_L)$ after the protocol and ρ are fixed but before the current distribution, target, oracle interaction, or tape. Proposition A.1 uses this same ordered catalog as the exact function-valued interface on actual canonical executions. Thus (A.37) uses only pre-instance catalog data and makes no selection depending on $\mathcal{D}$, h, a policy, a reply, or a tape. The map is deterministic, rather than sampled from a random feature-map law.

For each x, the binary catalog gives $\phi(x) \in \{-1, +1\}^L \subseteq \mathbb{R}^L$. By the Euclidean inner product and the coordinate definition,

$$\langle w, \phi(x) \rangle = \sum_{i=1}^{L} w_i \phi_i(x) = \sum_{i=1}^{L} w_i g_i(x),$$

with zero residual. Repeated functions simply give repeated legal coordinates and require no linear independence. For $L = 1$, the formula is $\langle (w_1), (g_1(x)) \rangle = w_1 g_1(x)$. The calculation is pointwise and finite-dimensional, so the cardinality of $\mathcal{X}$ is irrelevant. $\square$

Proposition A.5 (Exact target signs from the global simplex separator)**.** *Under Assumptions 1, 2, and 3, together with Proposition A.4 and Lemma A.11, the one fixed map ϕ satisfies*

$$\forall h \in \mathcal{H}\ \exists w_h \in \Delta_L\ \forall x \in \mathcal{X}, \qquad h(x)\langle w_h, \phi(x) \rangle \geq 1 - 2\varepsilon > \frac{1}{2} > 0. \tag{A.39}$$

Only h may affect the weight. The assertion includes empty $\mathcal{H}$, empty $\mathcal{X}$, $L = 1$, repeated coordinates, and infinite domains.

Proof. Assumption 1 gives $0 \leq \varepsilon < 1/4$, and therefore

$$2\varepsilon < \frac{1}{2} \quad \Longrightarrow \quad \rho_\varepsilon = 1 - 2\varepsilon > \frac{1}{2} > 0. \tag{A.40}$$

Fix $h \in \mathcal{H}$. Proposition A.4 gives $w_h \in \Delta_L$ such that simultaneously for every $x \in \mathcal{X}$,

$$h(x) \sum_{i=1}^{L} w_{h,i} g_i(x) \geq \rho_\varepsilon = 1 - 2\varepsilon. \tag{A.41}$$

Using (A.38) with $w = w_h$ yields

$$\begin{aligned} h(x)\langle w_h, \phi(x) \rangle &= h(x) \sum_{i=1}^{L} w_{h,i} g_i(x) \\ &\geq 1 - 2\varepsilon > \frac{1}{2} > 0. \end{aligned} \tag{A.42}$$

The equality is coordinate-for-coordinate, and the inequality is exactly (A.41); no approximate sign or distributional exceptional set appears.

The catalog and ϕ were fixed before the target was chosen. The proposition producing w_h fixes only h and uses the already fixed catalog; its conclusion contains no current distribution, policy, reply, or tape. The quantifier order is therefore one common ϕ, followed by a target-specific w_h, followed by all points. No measurable selector $h \mapsto w_h$ is required.

If $\mathcal{H} = \varnothing$, the target quantifier is vacuous. If $\mathcal{X} = \varnothing$ and $h \in \mathcal{H}$, Proposition A.4 supplies a point of nonempty Δ_L, and the pointwise clause is vacuous. If $L = 1$, membership in Δ_1 forces $w_h = (1)$. Duplicate coordinates do not change the exact identity, and infinite domains are covered pointwise by the global separator. $\square$

Proposition A.6 (Deterministic dimension closure with the catalog bound)**.** *Under Assumptions 1, 2, and 3, together with Proposition A.5 and the definition* (1.2),

$$\mathrm{dc}(\mathcal{H}) \leq L \leq B \left(1 + \frac{m}{\tau^2}\right)^k. \tag{A.43}$$

If $\mathcal{H} = \varnothing$ or $\mathcal{X} = \varnothing$, then $\mathrm{dc}(\mathcal{H}) = 0$. The specializations $m = 0$, $\varepsilon = 0$, $L = 1$, $B = 1$, $k = 1$, every $\tau > 0$, infinite domains, duplicate coordinates, and arbitrary including nonatomic tape spaces retain the literal conclusions.

Proof. Suppose first that $\mathcal{H} \neq \varnothing$ and $\mathcal{X} \neq \varnothing$. Proposition A.5 supplies one map $\phi : \mathcal{X} \to \mathbb{R}^L$ such that for every $h \in \mathcal{H}$ there is $w_h \in \mathbb{R}^L$ with

$$\forall x \in \mathcal{X}, \qquad h(x)\langle w_h, \phi(x)\rangle > 0.$$

Thus $d = L$, $\psi = \phi$, and $u_h = w_h$ are admissible in (1.2). Since $\mathrm{dc}(\mathcal{H})$ is the least admissible dimension,

$$\mathrm{dc}(\mathcal{H}) \leq L. \tag{A.44}$$

Now suppose $\mathcal{H} = \varnothing$ or $\mathcal{X} = \varnothing$. Let $d = 0$ and let $\psi_0 : \mathcal{X} \to \mathbb{R}^0$ be the unique map. If $\mathcal{H} = \varnothing$, the target quantifier in (1.2) is empty. If $\mathcal{X} = \varnothing$, choose the unique $u_h \in \mathbb{R}^0$ for each target; the point quantifier is empty. In either case the strict-sign condition is vacuous, so $d = 0$ is admissible. Since dimensions lie in $\mathbb{N}_0$,

$$\mathrm{dc}(\mathcal{H}) = 0. \tag{A.45}$$

As $L \geq 1$, this also implies (A.44) in the degenerate branches.

Assumption 3 states literally that

$$L \leq B\left(1 + \frac{m}{\tau^2}\right)^k. \tag{A.46}$$

Appending (A.46) to (A.44) proves (A.43). No term is dropped, dominated, or absorbed, and no hidden factor appears.

The boundary specializations are exact. If $m = 0$, then

$$L \leq B\left(1 + \frac{0}{\tau^2}\right)^k = B.$$

If $\varepsilon = 0$, the lower margin in (A.42) is one. Since $w_h \in \Delta_L$ and each $h(x)g_i(x) \in \{-1, +1\}$,

$$h(x)\langle w_h, \phi(x)\rangle \leq \sum_{i=1}^{L} w_{h,i} = 1,$$

so every nonvacuous margin is exactly one. If $L = 1$, then $\Delta_1 = \{(1)\}$ and $\phi(x) = (g_1(x))$. If $B = 1$, the catalog bound is $L \leq (1 + m/\tau^2)^k$; if $k = 1$, it is $L \leq B(1 + m/\tau^2)$. If $B = k = 1$ and $m = 0$, then $1 \leq L \leq 1$, so $L = 1$. Every $\tau > 0$ is allowed because $\tau^2 > 0$, and this argument imposes no further restriction.

The cardinality of $\mathcal{X}$ does not enter the coordinate identity, the pointwise margin, or the dimension implication. Repeated functions are legal coordinates, and no rank or distinctness premise is used. The conclusion is deterministic: the tape and its law do not occur in ϕ, w_h, or the final inequalities, so arbitrary and nonatomic tape spaces require no almost-sure, in-probability, or expectation qualification. □

Conclusion of the exact-coordinate and dimension argument. Proposition A.1 and Assumption 3 provide the exact ordered pre-instance catalog. Lemma A.11 uses those functions to define the common map and proves $\langle w_h, \phi(x)\rangle = \sum_i w_{h,i} g_i(x)$. For every target, Proposition A.4 supplies the same-catalog mixture at margin ρ_ε on every point. Proposition A.5 combines the two exact interfaces with $\varepsilon < 1/4$ to yield

$$\forall h \in \mathcal{H}\ \exists w_h \in \Delta_L\ \forall x \in \mathcal{X}, \qquad h(x)\langle w_h, \phi(x)\rangle \geq 1 - 2\varepsilon > \frac{1}{2} > 0.$$

Proposition A.6 applies the exact dimension definition and the literal catalog-size inequality to obtain

$$\mathrm{dc}(\mathcal{H}) \leq L \leq B\left(1 + \frac{m}{\tau^2}\right)^k.$$

The map is common and pre-instance, only the weight is target-specific, and the empty, zero-budget, zero-error, singleton-catalog, unit-constant, arbitrary-tolerance, infinite-domain, duplicate-coordinate, and arbitrary tape-space cases remain covered exactly. ☐

A.6 Proof of the Main Theorem

Proof of Theorem 3.1. Lemma A.1 and Proposition A.1 establish the legal canonical execution and its exact catalog membership. Lemma A.2 and Proposition A.2 turn the universal expected-risk guarantee into an exact per-distribution catalog-correlation witness. Proposition A.3, together with Lemma A.6, gives every finite simultaneous separator. Proposition A.4 globalizes those constraints in the fixed simplex. Finally, Lemma A.11 and Propositions A.5 and A.6 give exactly (3.1), (3.2), and the stated zero-dimensional empty cases. These are the conclusions of Theorem 3.1 under its three numbered assumptions. ☐